PROJECT REPORT

TITLE: Build a branch office network with 1 router, 1 switch, and configure 3 VLANs. Setup DHCP for all VLANs. Configure a default route to connect to headquarters. Add a DNS server. Use supernetting to summarize routes. Verify with network commands.

1. OBJECTIVES

- Design and configure a branch office network using VLANs.
- Implement inter-VLAN routing using the Router-on-a-Stick method.
- Configure DHCP on the router for automatic IP address allocation across all VLANs.
- Set up a DNS server for domain name resolution.
- Configure a static default route to connect the branch to headquarters.
- Apply route summarization (supernetting) to reduce routing table size.
- Verify network connectivity and correct operation using Cisco IOS show and ping commands.

2. THEORY

2.1 Virtual Local Area Network (VLAN)

A VLAN is a logical grouping of network devices that behave as if they are connected to the same broadcast domain, regardless of their physical location. By segmenting a network into multiple VLANs, broadcast traffic is contained, overall network performance is improved, and security between departments is strengthened. Communication between devices in different VLANs requires a Layer 3 device such as a router.

2.2 Inter-VLAN Routing — Router-on-a-Stick

Inter-VLAN routing allows traffic to flow between separate VLANs. In the Router-on-a-Stick model, a single physical router interface is logically subdivided into multiple sub-interfaces. Each sub-interface is configured with IEEE 802.1Q encapsulation and assigned the default gateway IP address for one VLAN, enabling the router to forward packets between VLANs over a single trunk link.

2.3 Dynamic Host Configuration Protocol (DHCP)

DHCP automatically assigns IP addresses and associated parameters — including subnet mask, default gateway, and DNS server — to client devices on request. Centralizing address management

on the router eliminates manual configuration, reduces the likelihood of addressing conflicts, and simplifies ongoing network administration.

2.4 Domain Name System (DNS)

DNS translates human-readable hostnames and fully qualified domain names into numeric IP addresses. A dedicated DNS server stores resource records and responds to client queries, allowing users to access network resources by name rather than by IP address.

2.5 Static Routing and Default Route

Static routing involves the manual entry of routes into a router's routing table. A default route (0.0.0.0/0) is a special static route that matches any destination for which no more specific route exists. In this lab, the default route forwards all unknown traffic toward the headquarters router, simulating external connectivity.

2.6 Route Summarization (Supernetting)

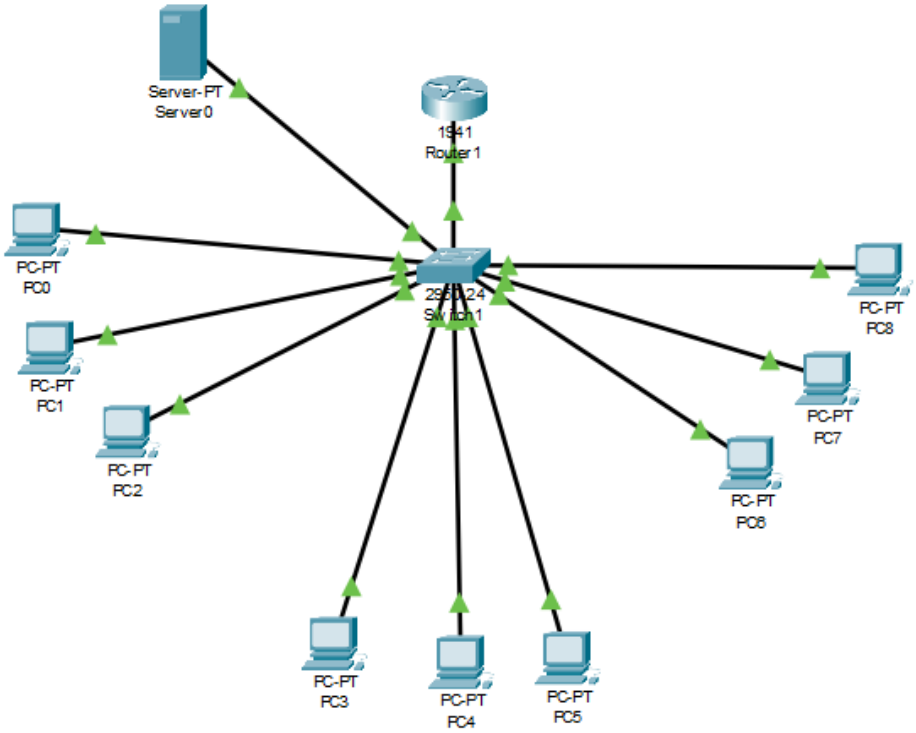
Route summarization (supernetting) aggregates multiple contiguous network prefixes into a single, less-specific route. This reduces the number of entries in the routing table, decreases routing protocol overhead, and improves convergence. In this lab, the three VLAN subnets (192.168.10.0/24, 192.168.20.0/24, and 192.168.30.0/24) are summarized into the single prefix 192.168.0.0/16 before being advertised toward headquarters.

2.7 Network Verification

Post-configuration verification confirms that all components are operating as intended. Cisco IOS commands such as show vlan brief, show ip route, show ip dhcp binding, and ping are used to validate VLAN assignment, routing table entries, DHCP lease allocation, and end-to-end connectivity respectively.

3. NETWORK TOPOLOGY

The branch office network consists of one router (R1), one Layer 2 switch (SW1), nine end-user PCs distributed across three VLANs, and one server providing DNS services. The router connects to the switch via a single trunk link and uses the Router-on-a-Stick configuration. A serial WAN interface on R1 provides connectivity to headquarters.



4. IP CONFIGURATION

4.1 Device Address Table

Device	Interface	IP Address	Subnet Mask	Gateway	DNS
Router R1	G0/0.10	192.168.10.1	255.255.255.0	N/A	N/A
Router R1	G0/0.20	192.168.20.1	255.255.255.0	N/A	N/A
Router R1	G0/0.30	192.168.30.1	255.255.255.0	N/A	N/A
Server0	Fa0	192.168.10.10	255.255.255.0	192.168.10.1	192.168.10.10
PCs	DHCP	Automatic	Automatic	Automatic	Automatic

4.2 VLAN Assignment Table

VLAN ID	Name	Switch Ports	Network	Default Gateway
10	SALES	Fa0/1 – Fa0/3	192.168.10.0 / 24	192.168.10.1
20	HR	Fa0/4 – Fa0/6	192.168.20.0 / 24	192.168.20.1
30	IT	Fa0/7 – Fa0/9	192.168.30.0 / 24	192.168.30.1

5. CONFIGURATION PROCEDURE

Step 1 — Switch Configuration

The switch was configured to create three VLANs and assign the appropriate access ports to each VLAN. The uplink port to the router was configured as a trunk, permitting all three VLANs. PortFast was enabled on access ports to reduce convergence delay for end devices.

```
enable
configure terminal

vlan 10
 name SALES
vlan 20
 name HR
vlan 30
 name IT
exit

interface range fa0/1 - 3
 switchport mode access
 switchport access vlan 10
 spanning-tree portfast
 no shutdown
exit

interface range fa0/4 - 6
 switchport mode access
 switchport access vlan 20
 spanning-tree portfast
 no shutdown
exit

interface range fa0/7 - 9
 switchport mode access
 switchport access vlan 30
 spanning-tree portfast
 no shutdown
exit
```

```
interface fa0/10
 switchport mode trunk
 switchport trunk allowed vlan 10,20,30
 no shutdown
exit

interface fa0/11
 switchport mode access
 switchport access vlan 10
 no shutdown
exit

end
write memory
```

Step 2 — Router Sub-Interface Configuration

The router's GigabitEthernet 0/0 physical interface was activated without an IP address. Three sub-interfaces were created, one for each VLAN. Each sub-interface was configured with 802.1Q encapsulation and assigned the default gateway IP address for its respective VLAN.

```
enable
configure terminal
hostname R1

interface g0/0
 no shutdown

interface g0/0.10
 encapsulation dot1Q 10
 ip address 192.168.10.1 255.255.255.0

interface g0/0.20
 encapsulation dot1Q 20
 ip address 192.168.20.1 255.255.255.0

interface g0/0.30
 encapsulation dot1Q 30
 ip address 192.168.30.1 255.255.255.0
exit

end
write memory
```

Step 3 — DHCP Configuration

The router was configured as a DHCP server for all three VLANs. Gateway addresses were excluded from the dynamic pools to prevent address conflicts. Each pool was assigned the correct network, default gateway, and DNS server address.

```
ip dhcp excluded-address 192.168.10.1
```

```
ip dhcp excluded-address 192.168.20.1
ip dhcp excluded-address 192.168.30.1

ip dhcp pool VLAN10
 network 192.168.10.0 255.255.255.0
 default-router 192.168.10.1
 dns-server 192.168.10.10
exit

ip dhcp pool VLAN20
 network 192.168.20.0 255.255.255.0
 default-router 192.168.20.1
 dns-server 192.168.10.10
exit

ip dhcp pool VLAN30
 network 192.168.30.0 255.255.255.0
 default-router 192.168.30.1
 dns-server 192.168.10.10
exit
```

Step 4 — Default Route Configuration

A default route was configured on R1 to forward all traffic destined for unknown networks through subinterface G0/0.10. This was used in Packet Tracer because no actual headquarters router exists in the topology.

```
ip route 0.0.0.0 0.0.0.0 g0/0.10
```

Step 5 — Route Summarization (Supernetting)

To reduce routing table entries advertised toward headquarters, the three /24 VLAN subnets were summarized into a single /16 prefix. This summary route covers all addresses in the 192.168.0.0 – 192.168.255.255 range and is forwarded via the WAN interface toward headquarters.

```
ip route 192.168.0.0 255.255.0.0 10.0.0.1
```

Step 6 — DNS Server Configuration

The server device in VLAN 10 was assigned a static IP address of 192.168.10.10 with a default gateway of 192.168.10.1. The DNS service was enabled and a resource record was added to map the hostname google.com to 8.8.8.8, enabling name resolution testing from end devices.

- Assigned static IP address: 192.168.10.10
- Subnet mask: 255.255.255.0
- Default gateway: 192.168.10.1
- DNS service enabled with record: google.com → 8.8.8.8

6. VERIFICATION

After completing all configuration steps, the following commands were executed to confirm correct operation of the network. Screenshots of the output are included below each command.

6.1 Show vlan brief

Confirms that VLANs 10, 20, and 30 are active and that the correct switch ports are assigned to each VLAN.

```
SW1# show vlan brief
```

```
VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/12, Fa0/13, Fa0/14, Fa0/15
                                                Fa0/16, Fa0/17, Fa0/18, Fa0/19
                                                Fa0/20, Fa0/21, Fa0/22, Fa0/23
                                                Fa0/24
10   SALES                            active    Fa0/1, Fa0/2, Fa0/3, Fa0/11
20   HR                               active    Fa0/4, Fa0/5, Fa0/6
30   IT                               active    Fa0/7, Fa0/8, Fa0/9
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
```

6.2 Show ip route

Displays the routing table on R1. Expected entries include connected routes for each sub-interface, the static default route (0.0.0.0/0), and the summary route (192.168.0.0/16).

```
R1# show ip route
```

```
R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 0.0.0.0 to network 0.0.0.0

     192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.10.0/24 is directly connected, GigabitEthernet0/0.10
L       192.168.10.1/32 is directly connected, GigabitEthernet0/0.10
     192.168.20.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.20.0/24 is directly connected, GigabitEthernet0/0.20
L       192.168.20.1/32 is directly connected, GigabitEthernet0/0.20
     192.168.30.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.30.0/24 is directly connected, GigabitEthernet0/0.30
L       192.168.30.1/32 is directly connected, GigabitEthernet0/0.30
S*   0.0.0.0/0 is directly connected, GigabitEthernet0/0.10
```

6.3 Show ip dhcp binding

Lists all DHCP leases currently issued by R1, confirming that end devices in each VLAN have received an IP address from the correct pool.

```
R1# show ip dhcp binding
```

```
IP address       Client-ID/              Lease expiration        Type
                 Hardware address
192.168.10.2     00D0.58A8.5C25           --                     Automatic
192.168.10.3     0060.70A8.8D34           --                     Automatic
192.168.10.4     00E0.F9A0.1EA0           --                     Automatic
192.168.20.2     0040.0B39.C5AD           --                     Automatic
192.168.20.3     00D0.FFD5.B830           --                     Automatic
192.168.20.4     00D0.BA71.966C           --                     Automatic
192.168.30.2     0004.9ABE.958A           --                     Automatic
192.168.30.3     00D0.BABA.4C2E           --                     Automatic
192.168.30.4     0002.4ABC.6905           --                     Automatic
```

6.4 Ping test

6.4.1 ping to VLAN 10 Gateway

This test verifies that the PC in VLAN 10 can successfully communicate with its default gateway.

```
PC> ping 192.168.10.1
```

```
C:\>ping 192.168.10.1

Pinging 192.168.10.1 with 32 bytes of data:

Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time=4ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.10.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 4ms, Average = 1ms
```

6.4.2 ping to VLAN 20 Gateway

This confirms that inter-VLAN routing is working, allowing devices in VLAN 10 to communicate with VLAN 20.

```
PC> ping 192.168.20.1
```

```
C:\>ping 192.168.20.1

Pinging 192.168.20.1 with 32 bytes of data:

Reply from 192.168.20.1: bytes=32 time<1ms TTL=255
Reply from 192.168.20.1: bytes=32 time=7ms TTL=255
Reply from 192.168.20.1: bytes=32 time<1ms TTL=255
Reply from 192.168.20.1: bytes=32 time=1ms TTL=255

Ping statistics for 192.168.20.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 7ms, Average = 2ms
```

6.4.3 ping to VLAN 30 Gateway

This ensures connectivity between VLAN 10 and VLAN 30 through the router.

```
PC> ping 192.168.30.1
```

```
C:\>ping 192.168.30.1

Pinging 192.168.30.1 with 32 bytes of data:

Reply from 192.168.30.1: bytes=32 time<1ms TTL=255
Reply from 192.168.30.1: bytes=32 time<1ms TTL=255
Reply from 192.168.30.1: bytes=32 time=1ms TTL=255
Reply from 192.168.30.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.30.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

6.4.4 Ping to Domain Name (DNS Test)

This test verifies DNS functionality. The domain name *google.com* is successfully resolved to its IP address (8.8.8.8), confirming that the DNS server is working correctly.

```
PC> ping google.com
```

```
C:\>ping google.com

Pinging 8.8.8.8 with 32 bytes of data:

Reply from 192.168.10.1: Destination host unreachable.
Reply from 192.168.10.1: Destination host unreachable.
Reply from 192.168.10.1: Destination host unreachable.
Reply from 192.168.10.1: Destination host unreachable.

Ping statistics for 8.8.8.8:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

The request shows “Destination host unreachable” because, although a default route to the headquarters network is configured, no HQ router exists in the topology. This behavior is expected.

7. RESULT

The branch office network was successfully configured and verified in Cisco Packet Tracer. All devices in each VLAN received IP addresses automatically via DHCP. Inter-VLAN communication was established through the Router-on-a-Stick configuration. DNS name resolution was confirmed, and the default route was successfully configured toward the simulated headquarters network. Route summarization consolidated the three VLAN subnets into a single advertised prefix, reducing routing overhead.

8. CONCLUSION

This project demonstrated the successful design and implementation of a branch office network using VLAN segmentation, inter-VLAN routing, DHCP, DNS, and static routing. VLANs improved network organization and security, while the Router-on-a-Stick method enabled efficient communication between different VLANs.

DHCP simplified IP address management by automating configuration, and DNS allowed user-friendly access through domain name resolution. Static routing and route summarization enhanced routing efficiency and scalability.

Overall, the project provided a strong understanding of essential networking concepts and their practical implementation using Cisco Packet Tracer.